

Critical reasoning

COMPLETE LEARNING GUIDE

What You'll Learn

Critical reasoning questions present a short argument — a set of facts (premises) leading to a conclusion — and ask you to identify its underlying assumption, strengthen or weaken it, spot its logical flaw, or distinguish what can be validly inferred from what the argument merely assumes. The core discipline is separating an argument into its explicit parts (premise, conclusion) and its implicit part (the unstated assumption bridging them), because most questions are really testing whether you can find that hidden bridge, not whether you agree with the conclusion. This sub-module covers argument structure, strengthen/weaken mechanics, the assumption-negation technique for verifying a true assumption, the inference-vs-assumption distinction, common logical flaws (especially correlation-vs-causation), bold-face role-identification questions, and cause-effect reasoning — each grounded in a fully worked example argument.

1. Argument Structure: Premise, Conclusion, Assumption

Every critical reasoning argument has stated premises (facts given), a conclusion (the claim being argued for), and at least one unstated assumption (a claim that must be true for the premises to actually support the conclusion) — identifying all three explicitly is the first step for any question type in this topic.

- 'Company X's sales increased after it launched a new ad campaign. Therefore, the ad campaign caused the sales increase.' Premise: sales rose after the campaign. Conclusion: the campaign caused the rise. Assumption: no other factor (seasonal demand, a competitor's price hike) caused the increase instead.

2. Strengthen and Weaken Questions

To weaken a causal argument, introduce a plausible alternative cause for the same outcome; to strengthen it, rule out alternative causes or provide direct evidence linking the stated cause to the effect.

- Weakens the ad-campaign argument: 'All companies in the sector, including competitors who ran no campaign, saw similar sales increases that quarter due to a holiday season.' Strengthens it: 'An A/B test showed customers exposed to the ad bought significantly more than a similar group not exposed to it.'

3. The Assumption-Negation Technique

To verify whether a candidate assumption is truly necessary for an argument, negate it (assume its opposite is true) — if the conclusion falls apart under the negation, the assumption was indeed necessary; if the conclusion still stands, it wasn't a required assumption.

- Negate 'no other factor caused the increase' !' 'another factor (seasonal demand) DID cause the increase.' Under this negation, the conclusion ('the campaign caused the increase') collapses — confirming this was indeed a necessary assumption of the original argument.

4. Inference vs Assumption: The Key Distinction

An assumption is an unstated premise the argument NEEDS to be valid (it comes before the conclusion, logically); an inference is a new conclusion that can be validly drawn FROM the stated facts (it comes after, as a further deduction) — these are frequently confused but test opposite directions of reasoning.

5. Identifying Logical Flaws

Common flaws include hasty generalization (concluding a universal rule from too few examples), circular reasoning (the conclusion restates a premise), and false cause (assuming correlation implies causation) — naming the specific flaw type, not just sensing something is wrong, is what the question is actually testing.

- 'Every swan I've seen is white, so all swans must be white.' Flaw: hasty generalization — a limited personal sample is treated as proof of a universal rule (and is factually wrong, since black swans exist).

6. Bold-Face and Evaluate-the-Argument Questions

Bold-face questions highlight two sentences within a passage and ask you to identify each one's logical ROLE (is it the main conclusion, a supporting premise, an opposing viewpoint the author rebuts, or background context) — read the whole passage first to establish its overall structure before assigning roles to the bolded parts.

7. Cause-Effect vs Mere Correlation

Two things happening together (correlation) does not prove one caused the other — a third, unstated factor may be driving both, and critical reasoning questions frequently test whether you can spot this specific flaw.

- 'Ice cream sales and drowning incidents both rise in summer.' This is correlation, not causation — the real common cause is hot weather, which independently increases both ice cream purchases and swimming (and therefore drowning risk).

Common Pitfalls

- Evaluating an argument's conclusion based on whether it seems true in the real world, rather than whether the given premises actually support it.
- Confusing an assumption (needed BEFORE the conclusion is valid) with an inference (a new claim drawn AFTER from the stated facts).
- Choosing a 'strengthen' option that merely restates the conclusion instead of one that rules out an alternative explanation.
- Weakening an argument with an irrelevant fact instead of one that introduces a genuine alternative cause for the same outcome.
- Treating correlation between two events as automatic proof that one caused the other.
- Skipping the assumption-negation check and guessing at an assumption option based on how 'reasonable' it sounds alone.

Key Takeaways

- Break every argument into premise(s), conclusion, and unstated assumption before answering any question type about it.
- To weaken a causal claim, introduce a plausible alternative cause; to strengthen it, rule out alternatives or add direct supporting evidence.
- Use the assumption-negation test: if negating a candidate assumption destroys the conclusion, it was a necessary assumption.
- Assumptions are needed BEFORE the conclusion follows; inferences are valid NEW claims drawn AFTER from the stated facts.
- Correlation is not causation — always consider whether a third factor could explain both observed events.
- Name the specific logical flaw (hasty generalization, false cause, circular reasoning) rather than just sensing the argument is weak.

Learn More

MindTools — Critical Thinking Skills

<https://www.mindtools.com/ao8m8mp/critical-thinking>

GeeksforGeeks — Verbal Ability Questions and Answers

<https://www.geeksforgeeks.org/verbal-ability-questions-and-answers/>